Power Prior for VARs

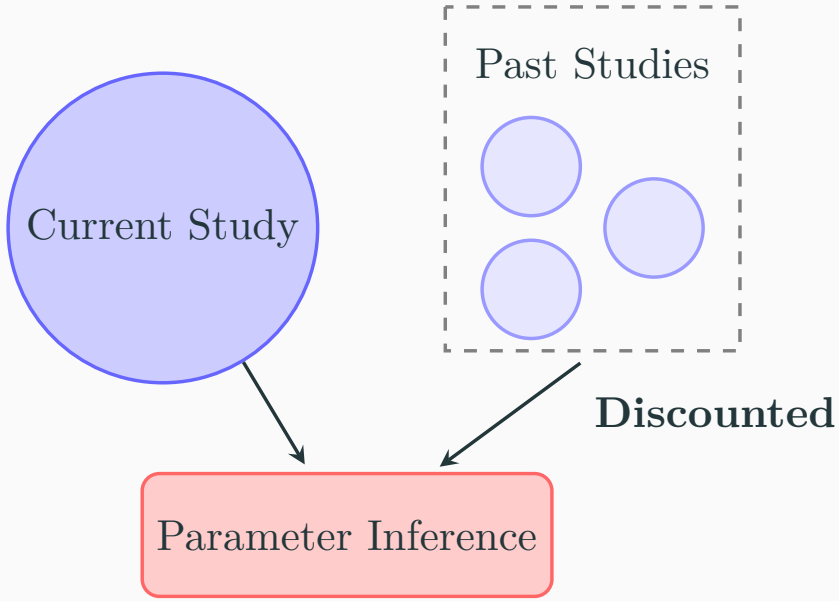
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October 28, 2026

About

How can we incorporate past/external data into our current data analysis? $\Rightarrow$ Popular tool in medical research: **power priors**!

For a review see Ibrahim et al. (2015) and Chen, Guan, et al. (2025)



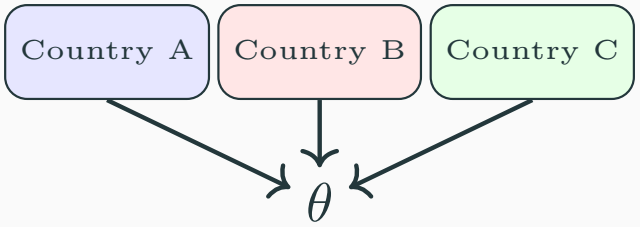
We adapt/extend power priors to VAR models!

(theory, simulations and applications)

How can we use this for VARs?

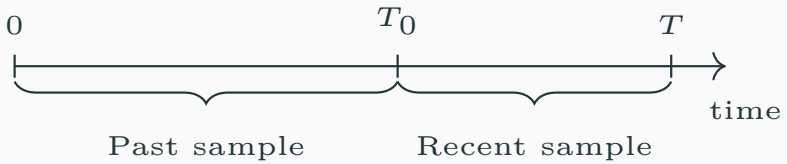
Use Different Groups

e.g. pool info to estimate long-run Phillips curve



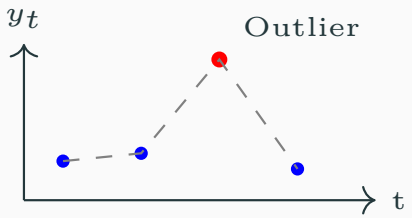
Past Sample Discounting

e.g. downweight earlier observations as in Baumeister and Hamilton (2019), Kiley (2024), etc.



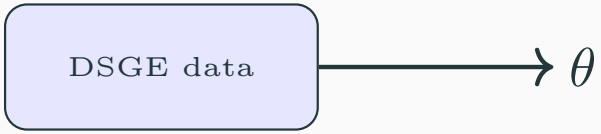
Sensitivity Analysis

e.g. discount Covid-Periods as in Lenza and Primiceri (2022), outliers: Billio et al. (2025)



Use Simulated Data

e.g. from DSGE-model as in Del Negro, Schorfheide, et al. (2007)



Developed theory unifies all of these applications!

how much to discount, country-selection, etc.

Focus on Cross-Country Information Borrowing

Multicountry models (Panel VARs, Global VARs, etc.), are widely applied to use or study cross-country relationships.

(common shocks, spillovers, etc.)

They model the **full joint distribution** of all countries, however,

often we are interested in only one country instead.

The **power prior** **allows borrowing information** and gives the **marginal country distribution**, keeping a familiar **single country VAR** structure.

$\Rightarrow$ **Conducting structural analysis as easy as for single country VARs!**

dimensions do not explode, no number of restrictions problem, using existing algorithms (Arias, Rubio-Ramirez, and Waggoner (2018)), etc.

Literature

Panel VAR / GVARs

see e.g. Korobilis (2016), Koop and Korobilis (2019), Angelini et al. (2019), Feldkircher et al. (2022), Pfarrhofer (2023), Prüser and Blagov (2026)

⇒ Middle ground between these and single country inference

Prior Choice for VARs

Doan, B. Litterman, and Sims (1984), R. Litterman (1986),Giannone, Lenza, and Primiceri (2015), Korobilis and Pettenuzzo (2019),George, Sun, and Ni (2008), Huber and Feldkircher (2019), Chan (2021), Chan (2022),Del Negro and Schorfheide (2004), Carriero, Clark, and Marcellino (2021), Giannone, Lenza, and Primiceri (2019), etc.

⇒ Power prior can be combined with all of them

Power Prior

Reviews in Ibrahim et al. (2015) and Chen, Guan, et al. (2025). Carvalho and Ibrahim (2021), Han, Ye, and Wang (2023),Banbeta, Lesaffre, and Rosmalen (2022),Ye et al. (2022),Chen and Ibrahim (2006), Shen et al. (2024)

⇒ Extend theory to VARs, new geometric results

Theory

Power Prior: Main Idea

Suppose we have access to **past/external data**, D_0, and we want to use this information for our **current data**, D_1, analysis. The power prior suggests:

$$\pi_{\text{post}}(\theta|D_0, D_1) \propto \underbrace{L(\theta|D_1)}_{\text{Current Likelihood}} \times \overbrace{\underbrace{L(\theta|D_0)^{\delta}}_{\text{Past Likelihood}} \times \underbrace{\pi_0(\theta)}_{\text{Baseline Prior}}}^{\text{Power Prior}}$$

- If $\delta = 0$: **No pooling, usual Bayes update**
- If $\delta = 1$: **Full pooling of information**

VAR Notation

Consider a VAR(p) written in the compact form

$$Y = ZA + U$$

- $Y = (y_1', \ldots, y_T')'$ (stacked $T \times n$ matrix)
- $Z = (x_1, \ldots, x_T)'$ with $x_t = (1, y_{t-1}', \ldots, y_{t-p}')$
- $A = (b', A_1', \ldots, A_p')'$

Assume $vec(U) \sim N(0, \Sigma \otimes I_T)$, thus the Likelihood is given by:

$$p(Y|A, \Sigma) = (2\pi)^{-\frac{Tn}{2}} |\Sigma|^{-\frac{T}{2}} exp(-\frac{1}{2} tr(\Sigma^{-1}(Y - ZA)'(Y - ZA)))$$

Denote by Y_0 the matrix Y for the past data and use Y_1 for the current data. Similar for Z.

Prior Specification

We construct the **normalized power prior** for different **baseline priors** $\pi_0(A, \Sigma, \delta)$ as

$$\pi_0(A, \Sigma, \delta|Y_0) = \frac{p(Y_0|A, \Sigma)^\delta \times \pi_0(A, \Sigma)}{\int\int p(Y_0|A, \Sigma)^\delta \times \pi_0(A, \Sigma) dA d\Sigma} \times \pi_0(\delta) \qquad (1)$$

Proposition 1

The normalized power prior, $\pi_0(A, \Sigma, \delta|Y_0)$, is proper if

$$\delta > C_{\pi_0}(T_0, n, p)$$

where $C_{\pi_0}(T_0, n, p) = 0$ whenever π_0 is proper

The normalization ensures that the Likelihood principle holds, see e.g. Duan (2005).

Power Posterior: Example

For a given δ, the posterior mean for a conjugate NIW baseline prior (with prior variance $\Omega(\lambda)$ and mean A_0) takes the following form:

$$\begin{aligned}\tilde{A} &= (\delta Z_0' Z_0 + Z_1' Z_1 + \Omega(\lambda)^{-1})^{-1}(\delta Z_0' Y_0 + Z_1' Y_1 + \Omega(\lambda)^{-1} A_0) \\ &= [\text{discount} \times \text{past data}] + [\text{current data}] + [\text{base prior}]\end{aligned}$$

This e.g. can implement **shrinkage towards external data** and/or **zero shrinkage**. similar to Prüser and Blagov (2026)

Alternative interpretation: δ inflates variance of D_0

$$\left[exp\left(-\frac{1}{2}\frac{(y_0 - x_0\beta)^2}{\sigma^2}\right)\right]^{\delta} = exp\left(-\frac{1}{2}\frac{(y_0 - x_0\beta)^2}{\frac{\sigma^2}{\delta}}\right)$$

Interpretation as Hierarchical Model

Consider a two-country hierarchical VAR:

$$y_{i,t} = X_{i,t}\beta_i + \epsilon_{i,t}, \qquad \epsilon_{i,t} \sim N(0, \Sigma_i), \qquad i \in \{A, B\},$$

where

$$X_{i,t} = I_n \otimes (1, y'_{i,t-1}, \ldots, y'_{i,t-p}), \qquad \beta_i = \text{vec}((b_i, A_{i,1}, \ldots, A_{i,p})').$$

Assume country-specific coefficients are drawn from a common distribution:

$$\beta_A \mid \bar{\beta}, \Omega \sim N(\bar{\beta}, \Omega), \qquad \beta_B \mid \bar{\beta}, \Omega \sim N(\bar{\beta}, \Omega), \qquad \pi(\bar{\beta}) \propto 1.$$

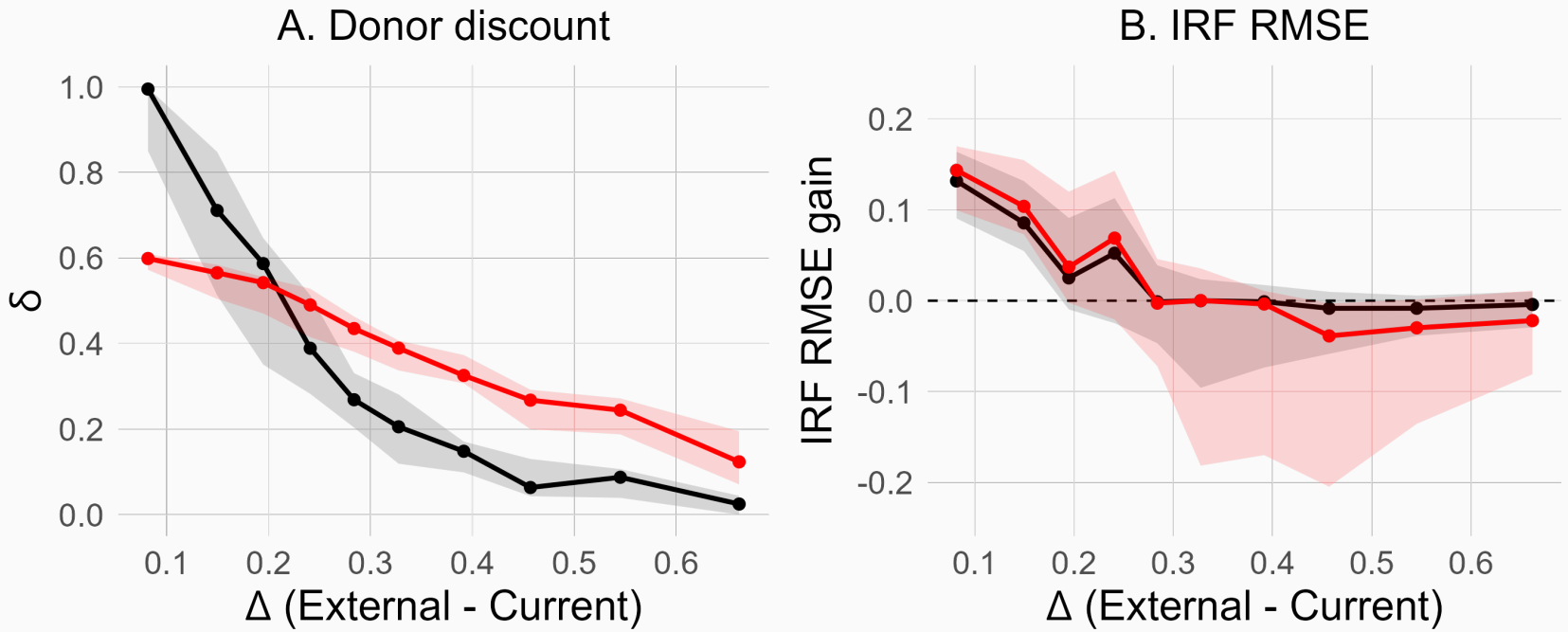
Proposition 2

Assume we apply the power prior for B using A as donor. Then, for a suitable mapping between Ω and δ,

$$p_{\text{PP}}(\beta_B \mid y_B, y_A) = p_{\text{Hier}}(\beta_B \mid y_B, y_A)$$

Estimation of Discount via Empirical Bayes

Simulation: 1000 different VAR(1) specifications for current and external data. Each dot are summary statistics for one parameter draw.

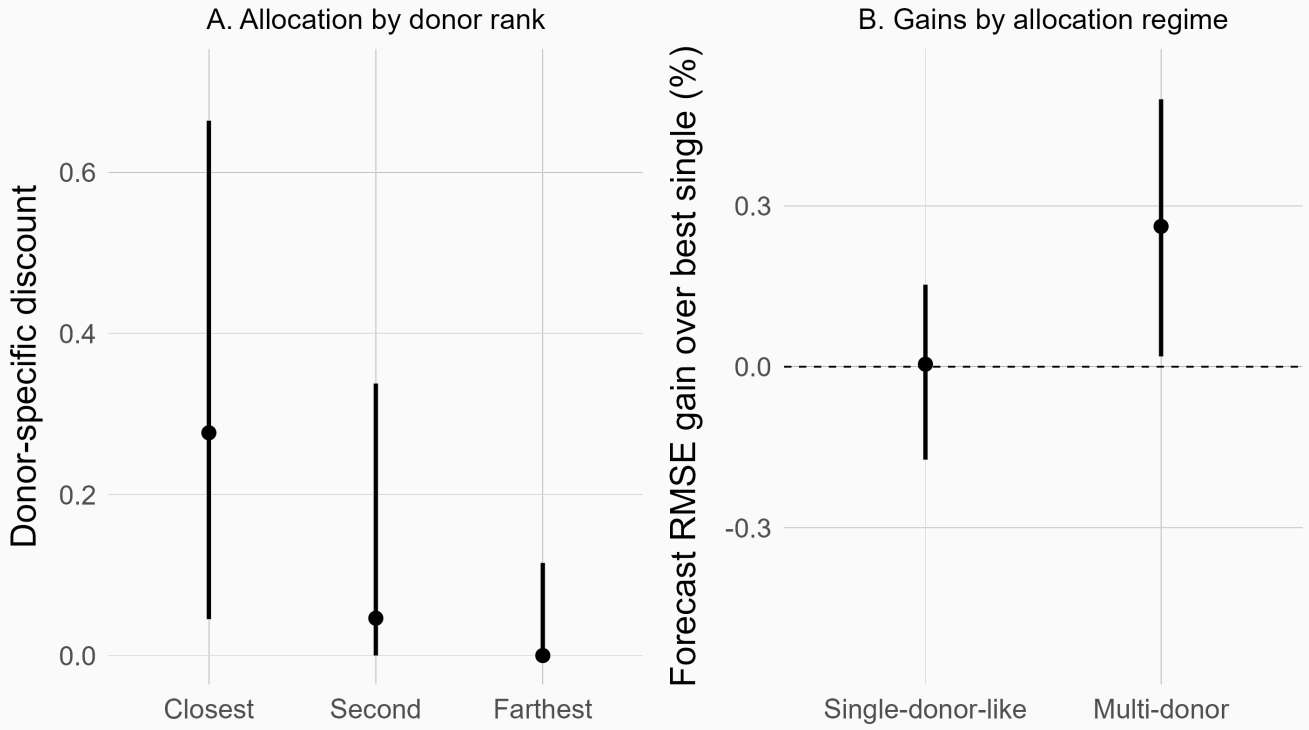


Red: Random discount with uniform prior, Black: Empirical Bayes

Frequentist Coverage

More than one Past Data Source: Hedging

In applications often we have many external data sources (countries). We show theoretically and in simulation that combinations of imperfect donors can help.



Hedging: The power prior might correct a bad baseline. Formal result

Application: Long-Run Phillips-curves

Motivation

Benati (2015) aims to find the **long-run PC** using a SVAR with **zero and sign restrictions**. As he notes:

- Estimating the long run is difficult
- Restrict for each country to period in which there was permanent variation in inflation!

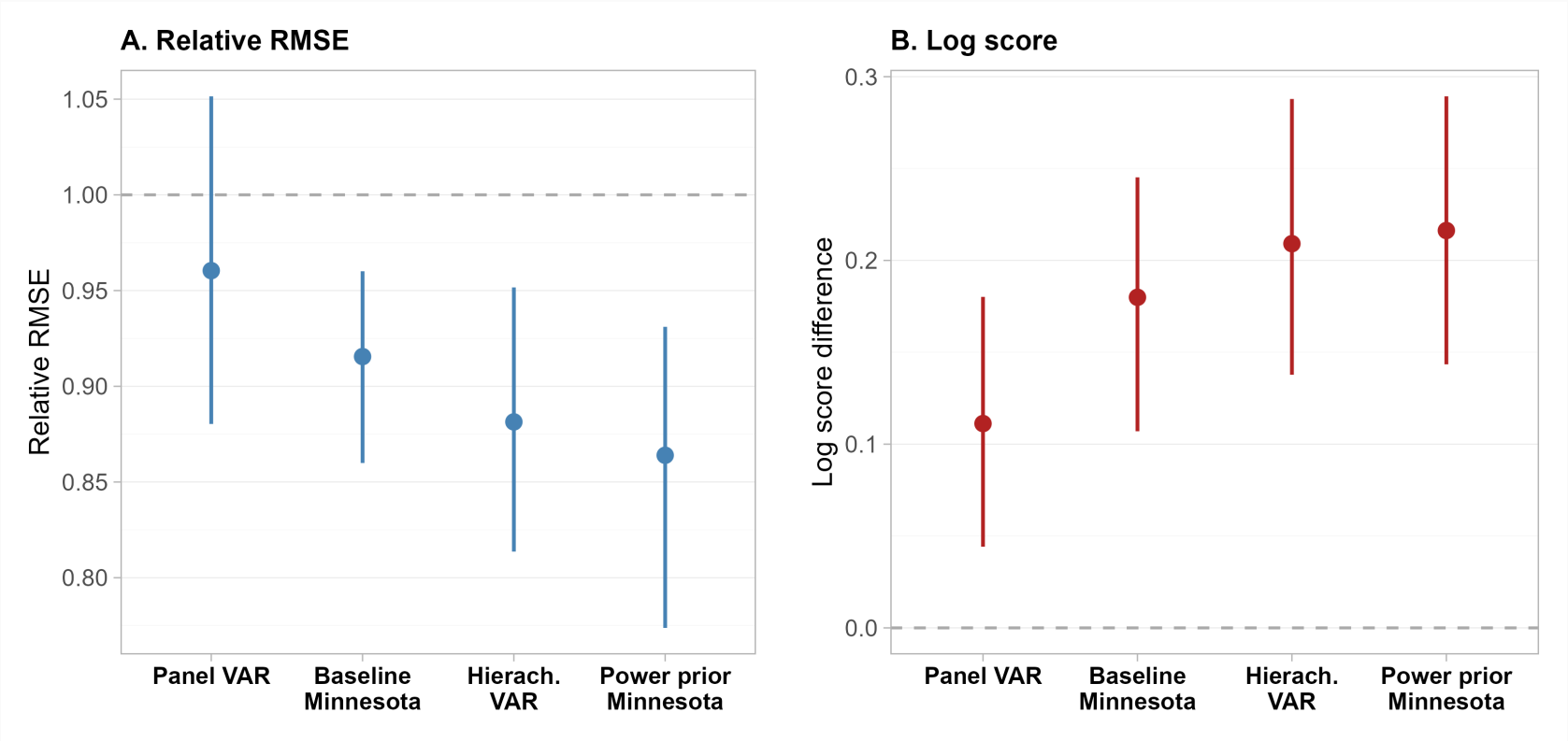
  US: 1959Q1-1979Q3, Euro Area: 1970Q1-1998Q4, UK: 1972Q1-1992Q3, Canada: 1961Q1-1990Q4, Australia: 1969Q3-1994Q2

The long-run PC is estimated separately for the US, UK, EU, CAN and AUS.

Keeping a diffuse baseline prior, what can we learn (partially) pooling information across countries?

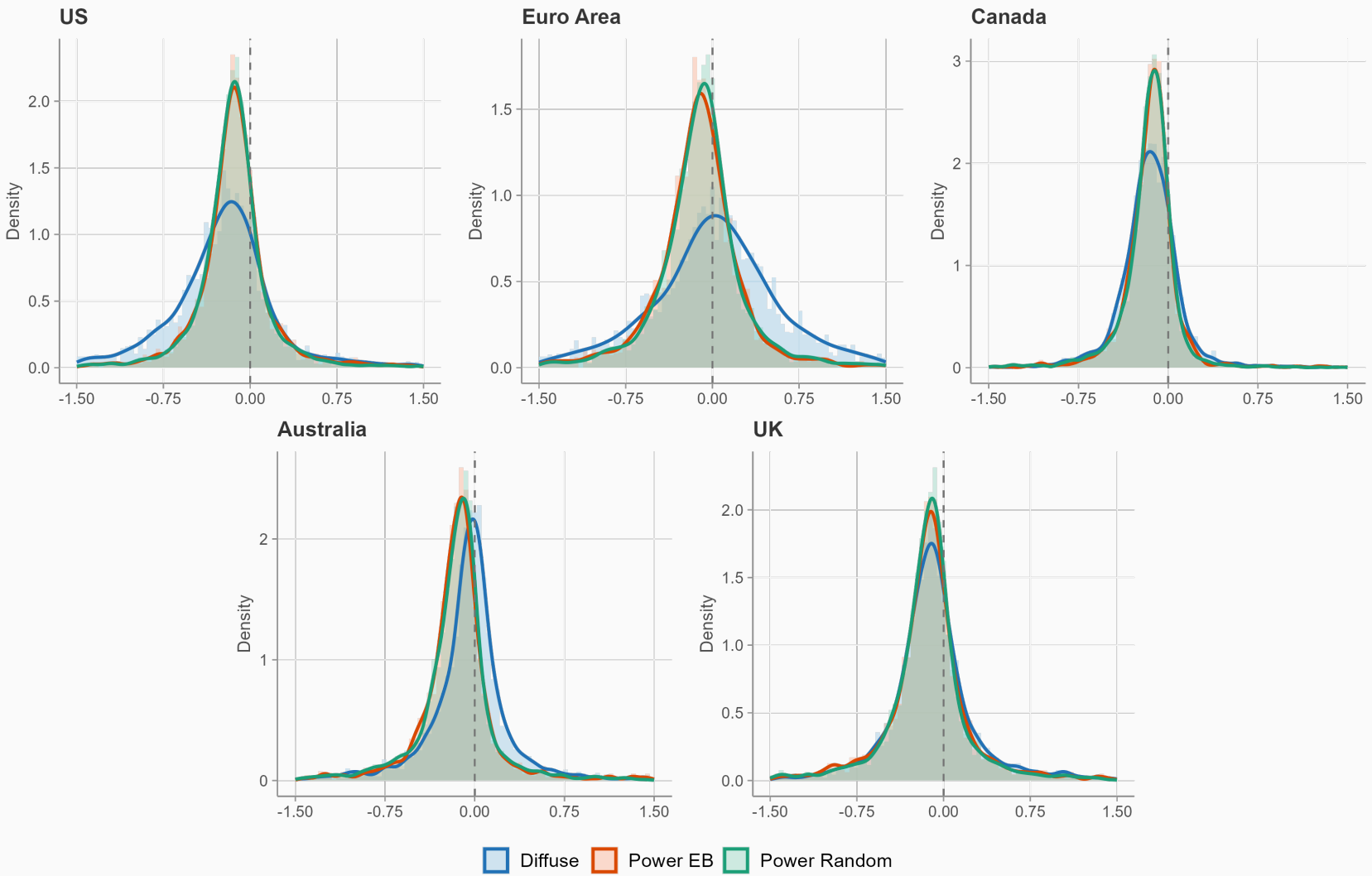
Forecasting with the Power Prior

Forecast performance out of sample 1985 Q1 to 2000 Q1. Left panel (< 1 better than baseline), Right panel (> 0 better than baseline) all relative to original diffuse prior.



Competitors are Minnesota variants including: Hierach VAR. (Prüser and Blagov (2026)), Panel VAR (Angelini et al. (2019))

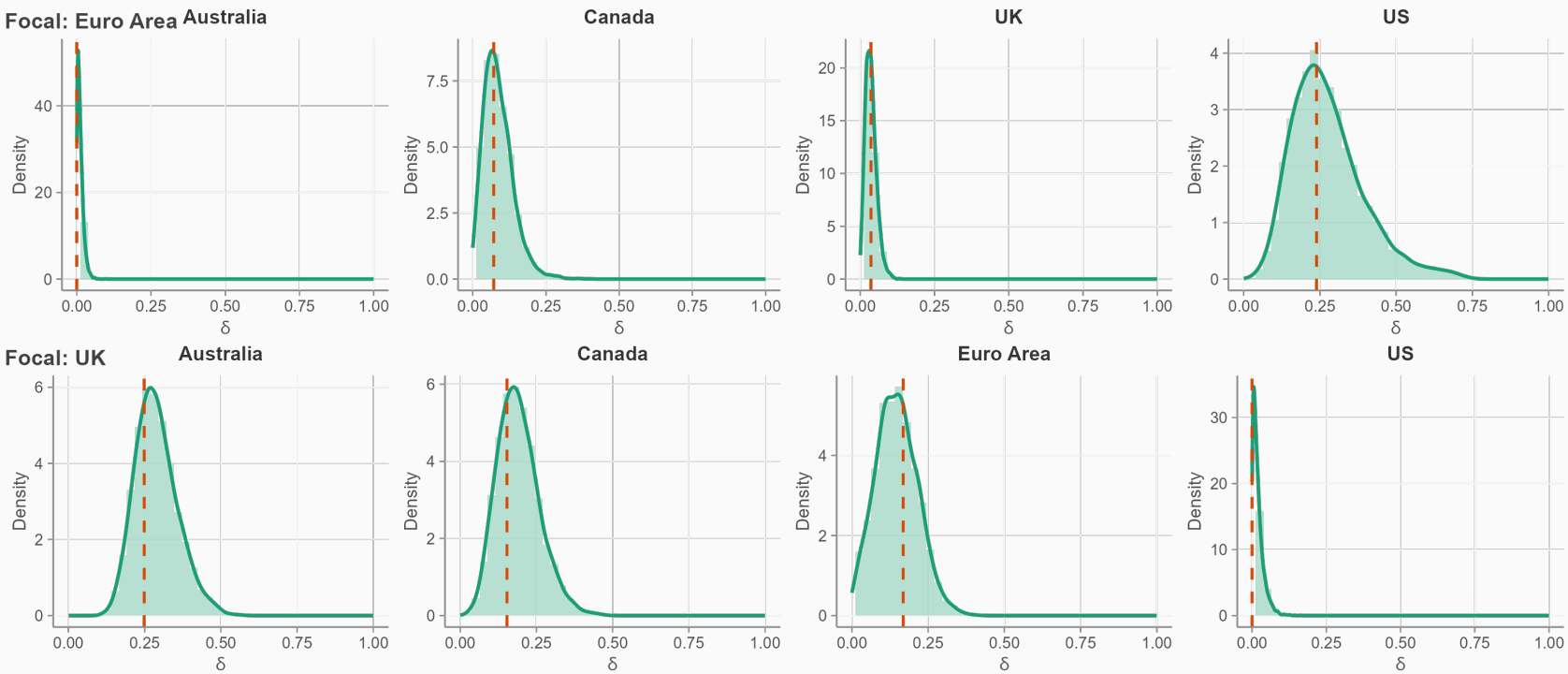
Longrun Effect of 1% Inflation Shock on Unemployment



This is still entirely data-driven! Details

Discount Factor Posterior: Examples

The framework flexibly selects from which country to borrow or not. Examples: Euro Area and UK and from which countries the models borrow.



Extensions

Asymmetric Prior and Variable Specific Discounting I

Chan (2022) starts with

$$A_0 y_t = b + B_1 y_{t-1} + \cdots + B_p y_{t-p} + \varepsilon_t$$

with A_0 unit lower-triangular, $\varepsilon_t \sim N(0, D)$ and $D = \text{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ Define $\tilde{w}_{i,t} = (-y_{1,t}, \ldots, -y_{i-1,t})$ and $\tilde{x}_t = (1, y'_{t-1}, \ldots, y'_{t-p})$, and set $x_{i,t} = (\tilde{w}_{i,t}, \tilde{x}_t)'$. Collect the equation-specific parameters in $\theta_i = (\beta'_i, \alpha'_i)'$.

Stacking over t yields, for each $i = 1, \ldots, n$,

$$y_i = X_i \theta_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma_i^2 I_T),$$

and because D is diagonal the likelihood factorizes:

$$p(y \mid \theta, \sigma^2) = \prod_{i=1}^{n} (2\pi\sigma_i^2)^{-\frac{T}{2}} \exp\left(-\frac{1}{2\sigma_i^2} (y_i - X_i\theta_i)'(y_i - X_i\theta_i) \right).$$

Asymmetric Prior and Variable Specific Discounting II

Similar as for the one discount case, the posterior mean is

$$\hat{K}_{\theta,i} = V_i^{-1} + \delta_i X_{0,i}' X_{0,i} + X_{1,i}' X_{1,i},$$
$$\hat{\theta}_i = \hat{K}_{\theta,i}^{-1}\left(V_i^{-1} m_i + \delta_i X_{0,i}' y_{0,i} + X_{1,i}' y_{1,i}\right),$$

Thus, we allow for discounts differing **variable-by-variable** and **country-by-country**.

Extended Forecasting Results

For the Phillips-Curve forecasting exercise, equation specific borrowing does not improve upon the country-discount approach but relative to its zero borrowing counterpart.

Model	Median RMSE ratio	RMSE win share	Median LPS difference	LPS win share	90% PI coverage
Power-prior variants					
Power prior, Minnesota baseline	0.864	32.5%	0.216	34.2%	96.3%
Power prior ACP	0.869	19.2%	0.235	28.7%	94.2%
Power prior, diffuse baseline	0.897	13.3%	0.147	4.2%	97.5%
Power-prior family	–	65.0%	–	67.1%	–
Single-country shrinkage benchmarks					
Baseline Minnesota	0.916	3.8%	0.180	8.3%	93.7%
Baseline ACP	0.901	8.3%	0.202	9.2%	93.7%
Multicountry benchmarks					
Hierach. VAR	0.881	17.9%	0.209	14.6%	94.4%
Panel VAR	0.960	5.0%	0.111	0.8%	94.6%

We additionally apply the power pior approach to the Mohaddes and Raissi (2024) dataset with 18 countries and 5 variables, i.e. 85 different discount factors, for which equation specific discounts matter more.

Conclusion

Conclusion

In this paper we ...

... **adapt the theory** of power priors to VARs

- conjugate results and properness
- relation to panel VARs
- geometric results for donor selection

... demonstrate its usefulness in **applications**

- structural PC curve estimation
- forecasting and posterior improvements
- validation using Mohaddes and Raissi (2024) dataset.

... extensive **Monte Carlo** simulations

- when does it work?
- how to deal with multiple external data sources

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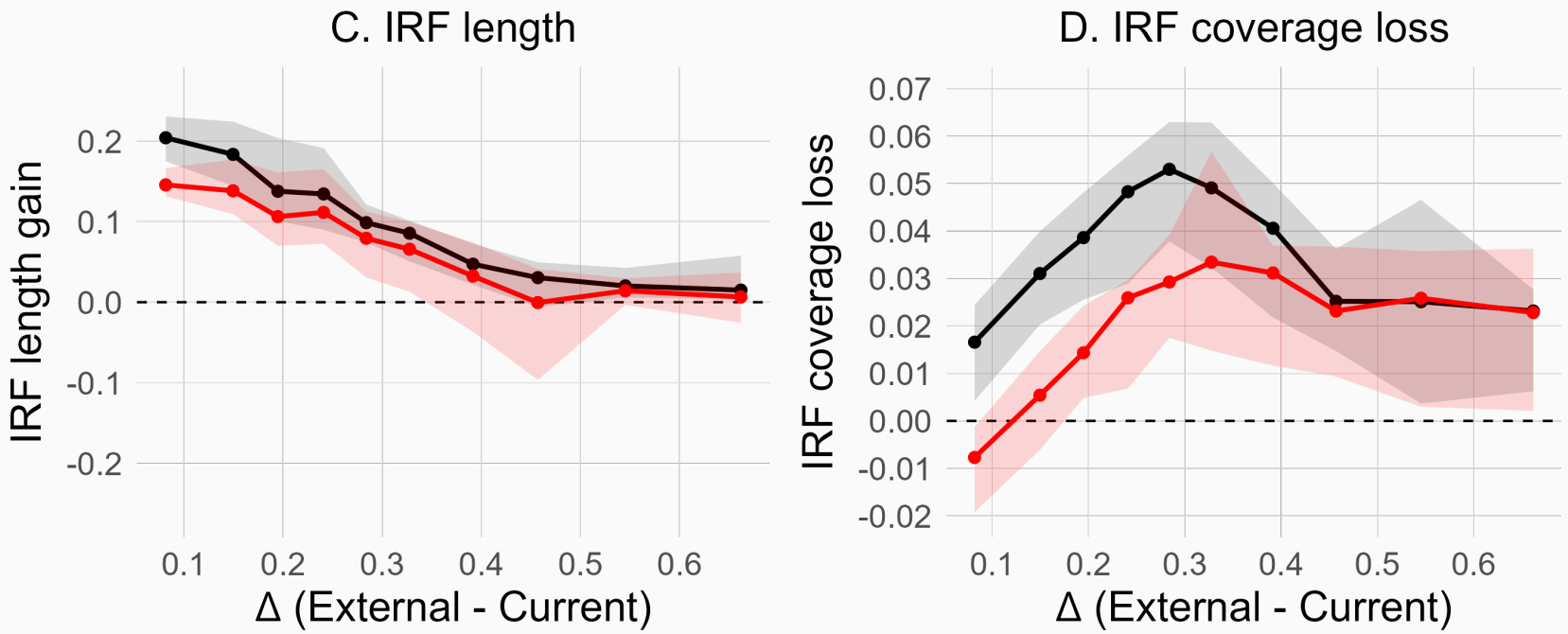
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Posterior Concentration and Frequentist Coverage



Red: Random discount with uniform prior, Black: Empirical Bayes

Back

Hedging Idea

Proposition 3

Let $y_i = \theta + \varepsilon_i$ with $\varepsilon_i \sim N(0, \sigma^2)$ and denote by D_0 and D_1 the external and current data, respectively. Let the oracle distribution be $\theta \sim \pi^* = N(\tau^*, \sigma^2)$. Let the baseline prior be $N(\tau, \sigma^2)$ and define the *normalized* power prior

$$\pi_\delta(\theta \mid D_0, \tau) \;=\; \frac{L(\theta \mid D_0)^\delta \, N(\theta; \tau, \sigma^2)}{\int L(\vartheta \mid D_0)^\delta \, N(\vartheta; \tau, \sigma^2) \, d\vartheta}, \qquad \delta \in [0, 1].$$

Denote by μ_0 the sample mean of D_0. If $\mu_0 < \tau^* < \tau$ or $\mu_0 > \tau^* > \tau$, then

$$\exists\, \delta \in (0, 1] \quad \text{s.t.} \quad \mathrm{KL}(\pi^* \parallel \pi_\delta(\cdot \mid D_0, \tau)) < \mathrm{KL}(\pi^* \parallel \pi_0(\cdot \mid D_0, \tau))\,,$$

i.e. $\delta = 0$ is suboptimal.

For multiple donors, what is important is the convex hull of the space spanned by all donors and the baseline prior. Back

Additional Results

Table 1: Long-run effect of a one percent monetary policy shock on unemployment

	Diffuse		Power EB			Power Random		
	$P(< 0)$	Median [5%, 95%]	$P(< 0)$	Median [5%, 95%]	CI reduction (%)	$P(< 0)$	Median [5%, 95%]	CI reduction (%)
Australia	0.579	-0.037 [-1.055, 0.624]	0.796	-0.140 [-0.846, 0.413]	25.0	0.790	-0.130 [-0.855, 0.434]	23.2
Canada	0.790	-0.149 [-0.600, 0.217]	0.841	-0.133 [-0.542, 0.163]	13.6	0.848	-0.135 [-0.586, 0.126]	12.7
Euro Area	0.467	0.042 [-2.403, 2.598]	0.663	-0.110 [-1.018, 0.755]	64.6	0.654	-0.093 [-0.961, 0.756]	65.7
UK	0.690	-0.113 [-0.972, 0.790]	0.731	-0.126 [-1.007, 0.640]	6.6	0.742	-0.124 [-0.937, 0.600]	12.8
US	0.747	-0.221 [-1.753, 0.938]	0.756	-0.138 [-0.757, 0.502]	53.2	0.775	-0.145 [-0.773, 0.401]	56.4

Back